

When Can We Trust Prediction Markets?

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Prediction Markets

Long tradition of thought

- Decentralized information centralizes through prices (Mises, 1920; Hayek, 1945)
- Early evaluations (Wolfers and Zitzewitz, 2006; Snowberg et al., 2013; Manski, 2006; Berg et al., 1997)
 - Limited by number of markets
 - Only able to compare to other measures or show reaction to single events

Recent explosion in prediction market availability and use

- Polymarket widely cited during 2024 election (Yaffe-Bellany and Griffith, 2024)
- US District Court legalizes betting on election outcomes (KalshiEX LLC and Commodity Futures Trading Commission, 2024)

Recent Papers Examine Newly Available Data

- Analysis of performance in 2024 election (Clinton and Huang, 2025)
 - Opportunity for arbitrage across platforms
 - Serial price correlation
 - Significant number of markets underperform random guesses (as measured by Brier score)
- Analysis of Polymarket trades (Reichenbach and Walther, 2025)
 - On average, good estimates of real world probability
 - Outcompetes bookmakers
 - Most traders lose money, growing share over time
 - ▶ Skilled traders exploit biases from unskilled
- Kalshi outcompetes traditional analysts in predicting macro indicators (Diercks et al., 2026)

This Paper

- Introduces new, marketwide metric of prediction accuracy
 - Measures systemic miscalibration of probability distributions
- Analyses patterns subsetting by category of market
 - Significant variation in prediction accuracy by category
- Finds weights on market characteristics that inform trustworthiness
 - Prediction markets much more useful when treated as a signal with strength informed by weight

Basic Theory: Supply/Demand

Agent j has endowment y_j and private belief q_j of an event occurring. They may buy or sell a security x with price π which pays \$1 if the event occurs. Assuming log preferences

$$\max_{x_j} EU_j = q_j \ln(y_j + (1 - \pi)x_j) + (1 - q_j) \ln(y_j - \pi x_j)$$

which yields demand/supply $x_j^* = y_j \frac{q_j - \pi}{\pi(1 - \pi)}$

Basic Theory: Market Clearing

Market clearing requires

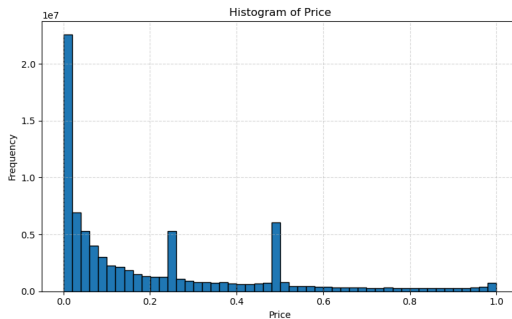
$$\int_0^{\pi} y \frac{q - \pi}{\pi(1 - \pi)} f(q, y) dq = \int_{\pi}^1 y \frac{\pi - q}{\pi(1 - \pi)} f(q, y) dq$$

which yields price $\pi = \frac{\int_0^1 qy f(q, y) dq}{\int_0^1 y f(q, y) dq}$

Such that prices are an average of beliefs weighted by income

Data

- Polymarket API provides minute-level price data for all markets in the last month
 - Prices defined as the midpoint of the spread or, in extreme cases, price of last trade
- Each market consists of two contracts, yes and no
 - Almost always exact opposites → only analyze yes
- Pull all market prices between April 15 and September 15, 2025
- Over 43,000 markets closed at this point, with 82.5 million price observations



Calibration Curve

- Moving average with prices on x-axis and outcome on y-axis^a
- Perfectly calibrated market would be a $y = x$ line
 - This doesn't mean perfect predictions, it means unbiased predictions
- Standard errors bootstrapped at the market level
- Area between the curve and the $y = x$ line is a measure of systemic bias

^aOutcomes are 0 or 1, average is percent of outcomes which occurred

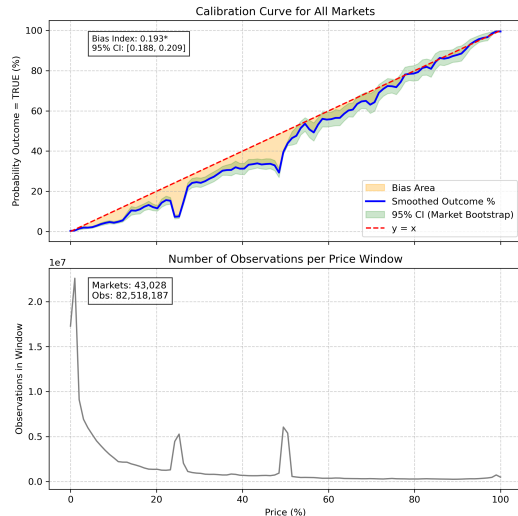


Figure: All Market Data, No Filters

Addressing Clustering

- Many observations at exactly 0, .25, .5, .75, 1.
 - Anchoring bias
 - Artifact of market creation
 - Wash trading

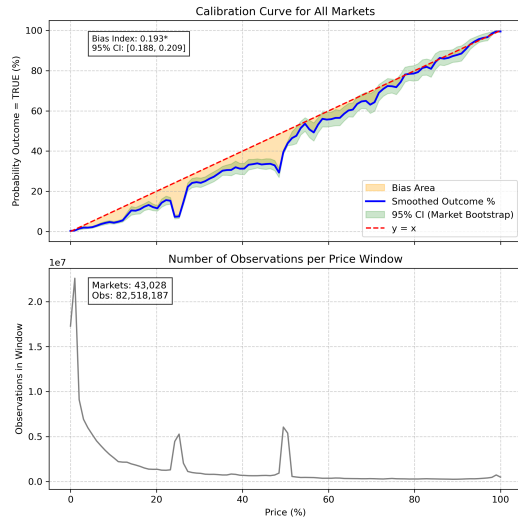


Figure: Filtering out Cluster Points

Inactivity

- Prices don't move in many markets for long periods
- This usually doesn't reflect confidence, it reflects inactivity
- Take advantage of price definition → trades almost always move the spread
- Remove repeated price observations

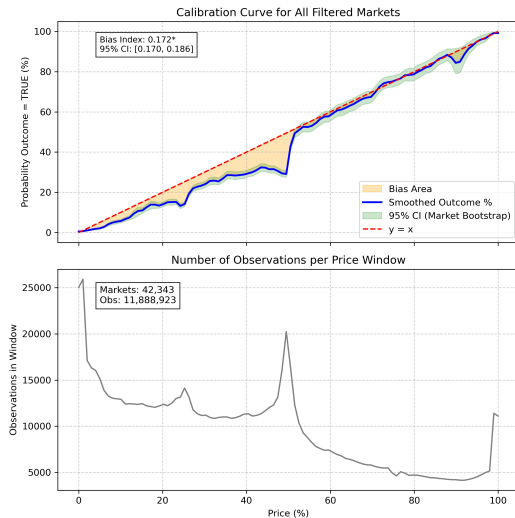


Figure: Final Data Calibration Curve

Categories

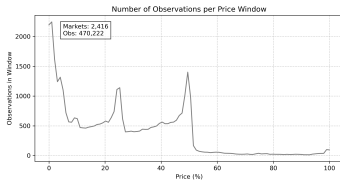
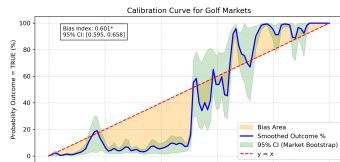
- All markets are tagged with relevant keyphrases
- Manually aggregate into broader categories ($\sim 2\%$ uncategorized)
- Split full dataset by category and graph calibration curves/bias indices
- Find remarkable heterogeneity.

Sports vs. Non-Sports

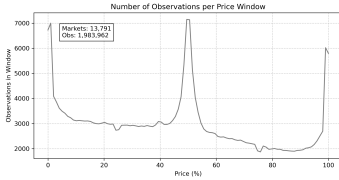
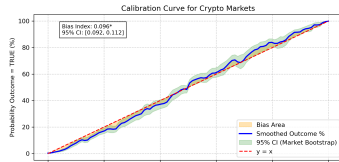
Category	Bias	Mkts	Obs
Soccer	0.222 (0.012)	3,342	496k
Tennis	0.346 (0.015)	1,591	207k
Baseball	0.372 (0.027)	2,683	706k
Hockey	0.377 (0.021)	445	121k
Basketball	0.407 (0.012)	2,742	544k
Football	0.441 (0.017)	1,621	639k
Golf	0.601 (0.016)	2,416	470k

Category	Bias	Mkts	Obs
Crypto	0.096 (0.005)	13,791	2.0M
Mentions	0.146 (0.008)	1,767	304k
All Filtered	0.172 (0.004)	42,343	11.9M
Domestic Politics	0.174 (0.011)	2,278	1.6M
Geopolitics	0.189 (0.015)	1,464	1.1M
Elections	0.189 (0.014)	1,460	960k
Weather	0.201 (0.012)	2,061	290k
Economy/Finance	0.230 (0.014)	1,423	1.3M
Culture	0.319 (0.019)	2,790	1.9M

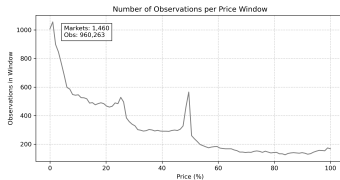
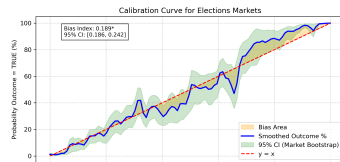
Selected Examples



Golf



Crypto



Elections

Convergence Over Time

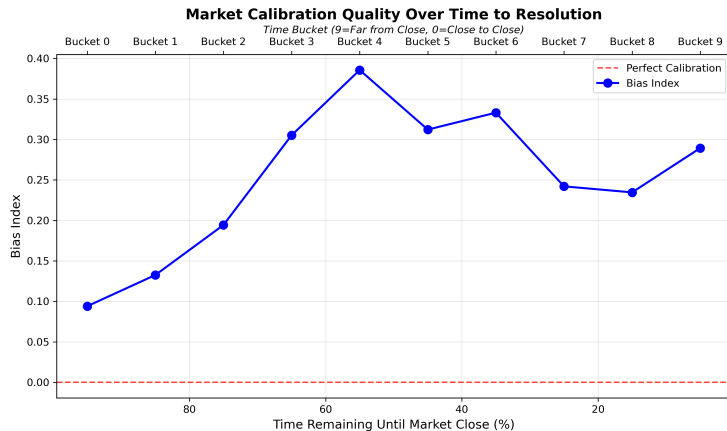


Figure: Bias Reduces as we Near Market Close

Convergence over Volume

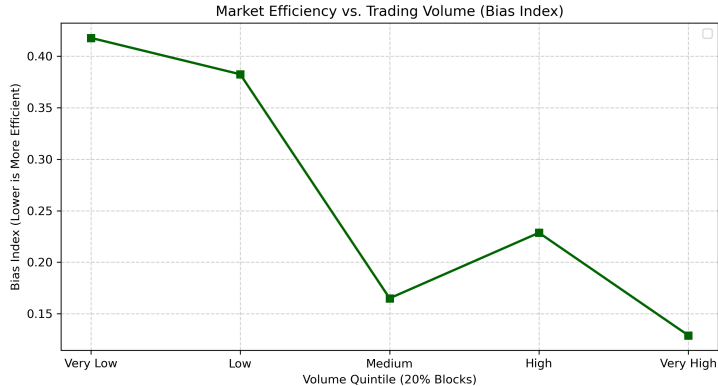


Figure: Bias Reduces in Trading Volume

Correction

- So the signal isn't just noisy, it's biased in predictable ways. Can we do better?
- Train a random forest to predict outcome from observations about the market
- Train on 80% of markets, evaluate on the remaining 20%

What is a Random Forest?

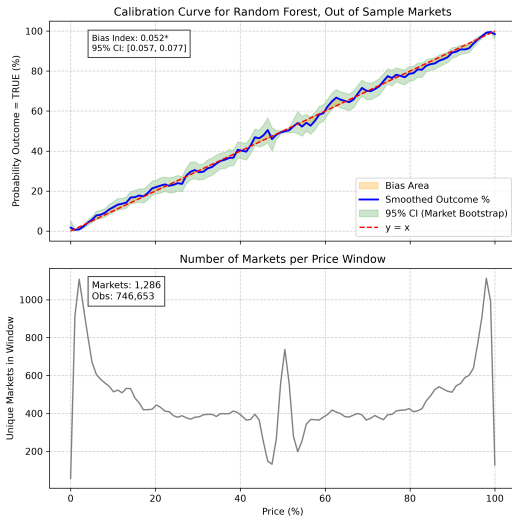
- An ensemble of many decision trees (~ 200 here)
- Each tree repeatedly splits the data on a threshold (e.g. “is price $< 15\%$?”) to separate markets that resolved true from those that didn't
 - Because later splits are nested inside earlier ones, trees capture interactions automatically
 - The effect of volume can depend on where price sits
- Each tree trains on a random resample of markets and, at every split, is only allowed to consider a random subset of variables
 - Decorrelates the trees, so each one's individual errors average out across the forest rather than compounding
- Prediction = average, across all 200 trees, of their predicted outcome

Correction Results

Cuts miscalibration by 70% out of sample

Bias Index: 0.172 $\rightarrow$ 0.052

Variable	Importance
Price	86.5%
Volume (log, sq)	5.3%
Volume (log)	5.1%
Log Time Sq	1.3%
Log Time	1.2%
Is Sports	0.6%



Conclusion

- Prediction markets appear far better calibrated when taken as a whole than looking at any given slice
- Systemic, heterogenous miscalibration
- Massive opportunity for arbitrage. Why not?
 - Too much "dumb money"
 - Time horizon for certain markets
 - Only trading fees on very short-term crypto markets
 - Friction to opening crypto wallet?
 - Systemic analysis not yet done?
- Able to re-weight how much we trust any one market to be well-calibrated on average

Citations

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